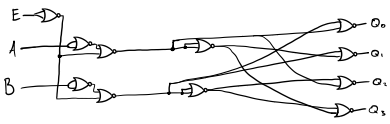


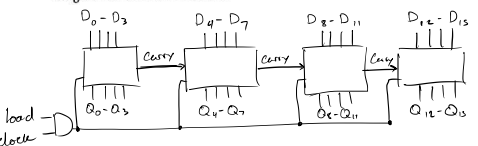
2-4. Draw the logic diagram of a 2-to-4-line decoder with only NOR gates. Include an enable input E .



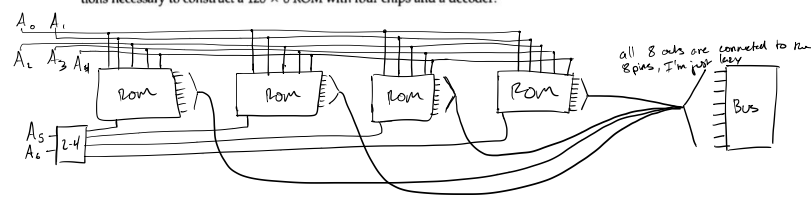
2-10. What is the purpose of the buffer gate in the clock input of the register of Fig. 2-7?

Strongly means the clock signal ensures integrity of the circuit

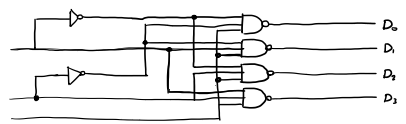
2-17. Show the connections between four 4-bit binary counters with parallel load (Fig. 2-11) to produce a 16-bit binary counter with parallel load. Use a block diagram for each 4-bit counter.



2-22. Given a 32×8 ROM chip with an enable input, show the external connections necessary to construct a 128×8 ROM with four chips and a decoder.



2-5. Modify the decoder of Fig. 2-2 so that the circuit is enabled when $E = 1$ and disabled when $E = 0$. List the modified truth table.



2-12. The content of a 4-bit register is initially 1101. The register is shifted six times to the right with the serial input being 101101. What is the content of the register after each shift?

Shift 1 (input=1) $\rightarrow$ 1110
 Shift 2 (input=0) $\rightarrow$ 0111
 Shift 3 (input=1) $\rightarrow$ 1011
 Shift 4 (input=0) $\rightarrow$ 0101
 Shift 5 (input=1) $\rightarrow$ 1011
 Shift 6 (input=0) $\rightarrow$ 0101

2-19. The following memory units are specified by the number of words times the number of bits per word. How many address lines and input-output data lines are needed in each case? (a) $2K \times 16$; (b) $64K \times 8$; (c) $16M \times 32$; (d) $4G \times 64$.

a) 11 address lines, 16 data lines
 b) 16 address lines, 8 data lines
 c) 24 address lines, 32 data lines
 d) 32 address lines, 64 data lines

2-9. Include a two-input AND gate with the register of Fig. 2-6 and connect the gate output to the clock inputs of all the flip-flops. One input of the AND gate receives the clock pulses from the clock pulse generator. The other input of the AND gate provides a parallel load control. Explain the operation of the modified register.

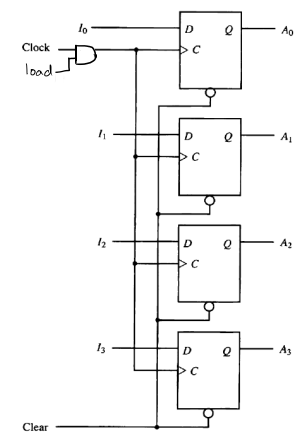


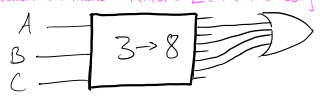
Figure 2-6 4-bit register.

9. Design the logic functions A_1 and A_0 for:

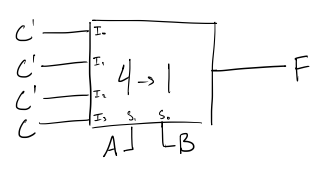
Inputs				Outputs		
D_3	D_2	D_1	D_0	A_1	A_0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

• $A_1 = D_3 + D_0$ • $A_0 = D_3 + D_2 + D_1$

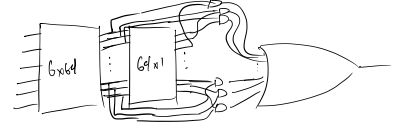
12. Implement the function $F(A, B, C) = \sum(1, 3, 5, 6)$ using a decoder



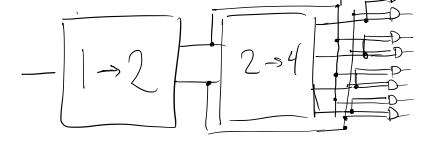
13. Implement the 3-variable function $F(A, B, C) = \sum(0, 2, 4, 7)$ with a 4x1 multiplexer.



10. Design a 64x1 multiplexer with a 6x4 decoder and "And" and "Or" gates



11. Design a 3-to-6 line decoder with a 2-to-4 line decoder, a 1-to-2 line decoder, 3 five 2-input And gates, and that all decoder outputs are used at least once.

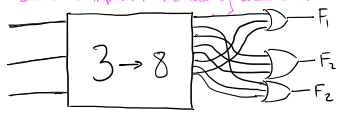


13. Design the combinational circuit to implement the following boolean function:

$$F_1 = x'y'z' + xz$$

$$F_2 = xy'z' + x'y$$

$$F_3 = x'y'z + xy$$



14. Tabulate the truth table for a 8x4 ROM that implements the following functions:

$$A(X, Y, Z) = \sum(0, 1, 2, 6, 7)$$

$$B(X, Y, Z) = \sum(2, 3, 4, 5, 6)$$

$$C(X, Y, Z) = \sum(2, 6)$$

$$D(X, Y, Z) = \sum(1, 2, 3, 5, 6, 7)$$

XYZ	A	B	C	D
000	1	0	0	0
001	1	0	0	1
010	1	1	1	1
011	0	1	0	1
100	0	1	0	0
101	0	1	0	1
110	1	1	1	1
111	1	0	0	1